

CHAMPION AGDM, AB, Canada

WMO: 712170

Lat: 50.2953N

Lon: 113.3467W

Elev: 995

StdP: 89.93

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.6	-25.4	-31.9	0.2	-28.4	-28.4	0.3	-25.3	13.2	2.3	11.8	0.9	2.5	90	0.707

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	30.8	15.9	28.7	15.5	26.8	14.9	18.2	25.8	17.1	25.0	16.1	24.4	4.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.6	12.5	21.4	14.2	11.4	19.8	13.1	10.6	18.3	55.6	25.9	51.8	24.9	48.7	24.5	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	10.7	9.2	-32.7	34.1	3.5	2.4	-35.2	35.8	-37.3	37.2	-39.2	38.6	-41.8	40.3		
(5)			-32.8	20.3	3.5	1.3	-35.3	21.3	-37.4	22.1	-39.3	22.8	-41.8	23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-6.6	-7.4	-1.2	4.5	10.2	14.1	17.4	16.6	11.8	5.2	-2.1	-7.5	(6)
(7)		DBStd	10.85	8.47	8.05	7.40	4.90	4.42	3.07	2.90	3.29	4.61	5.35	7.61	8.09	(7)
(8)		HDD10.0	2682	516	488	347	172	52	3	0	1	33	162	364	542	(8)
(9)		HDD18.3	5042	774	721	605	415	253	131	51	73	199	406	613	801	(9)
(10)		CDD10.0	733	0	0	1	7	59	126	231	206	87	14	2	0	(10)
(11)		CDD18.3	51	0	0	0	0	1	4	23	19	3	0	0	0	(11)
(12)		CDH23.3	1437	0	0	1	7	78	124	533	514	173	8	0	0	(12)
(13)		CDH26.7	437	0	0	0	1	14	26	172	174	50	1	0	0	(13)
(14)	Wind	WSAvg	4.1	4.2	3.7	4.4	4.6	4.3	3.6	3.6	3.9	4.3	4.3	3.9	(14)	
(15)	Precipitation	PrecAvg	408	10	13	20	27	60	100	50	41	38	18	16	12	(15)
(16)		PrecMax	658	22	36	47	52	109	289	120	94	138	43	44	29	(16)
(17)		PrecMin	179	1	0	4	9	8	44	8	2	5	4	1	0	(17)
(18)		PrecStd	107	6	10	13	12	32	59	35	23	32	10	11	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	12.3	17.4	23.7	28.5	30.0	33.2	33.1	30.9	24.1	18.0	11.0	(19)
(20)			MCWB	5.5	5.0	7.5	10.4	13.4	16.6	16.5	15.7	14.4	11.2	8.8	5.1	(20)
(21)		2%	DB	8.2	8.3	14.3	19.9	25.7	26.6	30.8	30.7	28.1	20.7	13.2	7.9	(21)
(22)			MCWB	3.5	3.0	6.2	8.9	12.2	14.7	16.9	15.4	13.7	10.4	6.3	3.2	(22)
(23)		5%	DB	6.1	5.9	11.9	17.0	22.7	24.3	28.5	28.5	25.0	17.5	10.5	5.7	(23)
(24)			MCWB	2.3	1.7	4.9	7.4	11.2	14.4	16.5	15.2	12.9	9.0	5.0	1.9	(24)
(25)		10%	DB	4.4	3.5	9.2	14.0	20.1	22.3	26.4	26.3	22.0	14.6	7.8	3.7	(25)
(26)			MCWB	1.3	0.4	3.8	5.8	10.1	13.1	16.2	14.7	11.8	7.4	3.2	0.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	5.7	8.5	11.0	14.2	18.4	20.9	18.2	15.5	12.5	9.2	5.7	(27)
(28)			MCDWB	10.2	11.3	15.8	22.6	25.0	25.2	26.9	27.0	26.2	22.0	16.5	9.8	(28)
(29)		2%	WB	3.8	3.4	6.7	9.3	12.9	16.6	19.0	17.0	14.2	10.9	7.0	3.5	(29)
(30)			MCDWB	7.9	8.1	13.3	18.4	23.6	24.2	26.6	26.6	25.5	19.3	12.6	7.3	(30)
(31)		5%	WB	2.6	1.8	5.2	7.8	11.8	15.1	17.9	16.1	13.2	9.3	5.2	2.1	(31)
(32)			MCDWB	5.8	5.4	11.1	16.1	21.2	22.0	25.3	25.6	23.8	16.7	10.2	5.4	(32)
(33)		10%	WB	1.4	0.4	3.8	6.4	10.6	14.0	16.9	15.2	12.1	7.9	3.6	0.5	(33)
(34)			MCDWB	4.3	3.4	9.1	13.2	18.7	20.7	24.2	24.3	21.2	13.9	7.4	3.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.8	12.3	12.4	13.7	13.8	13.2	15.8	16.2	15.3	14.1	12.3	11.4	(35)
(36)			MCDBR	12.3	15.2	16.4	19.8	19.0	17.5	19.8	21.0	21.3	18.7	15.2	12.6	(36)
(37)			MCWBR	8.5	10.4	9.4	9.9	8.1	7.6	8.2	8.0	8.8	9.6	9.0	8.6	(37)
(38)		5% WB	MCDWR	12.2	14.7	15.6	18.4	17.1	15.4	16.7	18.3	19.7	17.9	14.8	12.5	(38)
(39)			MCWBR	8.6	10.3	9.3	9.4	7.5	7.4	8.2	7.5	8.3	9.5	9.0	8.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.254	0.262	0.285	0.325	0.338	0.334	0.337	0.352	0.313	0.281	0.258	0.234	(40)	
(41)		taud	2.427	2.473	2.464	2.366	2.384	2.440	2.447	2.388	2.475	2.532	2.489	2.438	(41)	
(42)		Ebn at Noon	812	901	932	920	919	922	914	880	887	864	804	791	(42)	
(43)		Edh at Noon	59	73	89	110	113	108	106	107	88	68	55	50	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.22	2.21	3.52	4.83	5.77	6.34	6.92	5.52	3.91	2.42	1.33	0.93	(44)	
(45)		RadStd	0.10	0.20	0.27	0.33	0.43	0.41	0.34	0.35	0.35	0.27	0.12	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page